STM Exercise 5

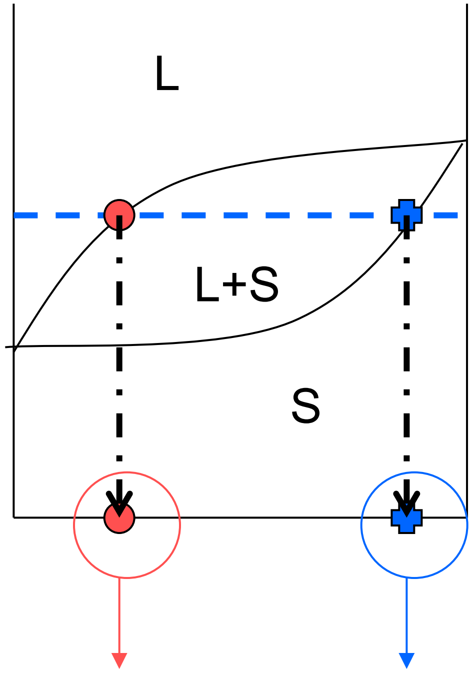
Phase Diagrams

(full solubility in the solid state)

(no solubility in the solid state)

Determination of the compositions of phases in equilibrium in a biphasic zone:

HORIZONTAL RULE



Liquid composition

Solid composition

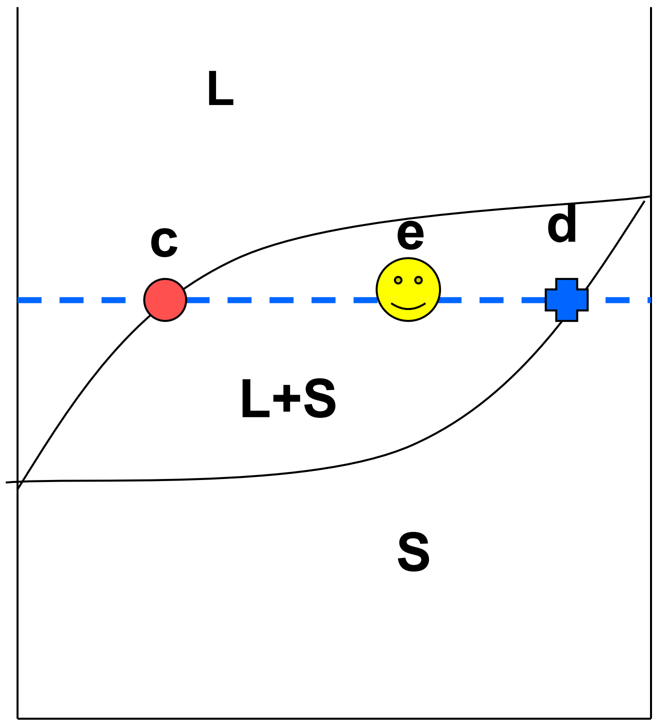
- Draw the tie line - - - - - -
- Individuate intersections with **liquidus** (•) and **solidus** (✚) lines
- The **composition of the liquid** is the value on the x axix correspondent to the intersection between the tie line and the liquidus line.
- The **composition of the solid** is the value on the x axis correspondent to the intersection of the tie line with the solidus line.

LIQUIDUS CURVE = **above** it **only liquid** is present

SOLIDUS CURVE = **below** it **only solid** is present

Phases relative amounts determination in a biphasic zone, at equilibrium:

LEVER RULE



- Draw the tie line - - - - - -
- Individuate the intersection between tie line and liquidus line (● c) and solidus line (✚ **d**)
- The % of phases in a point of the biphasic region (e) is defined as:

%L = (ed/cd)*100

%S = (ce/cd)*100

The lever rule

If an alloy consists of more than one phase, the amount of each phase present can be found by applying **the lever rule** to the phase diagram.

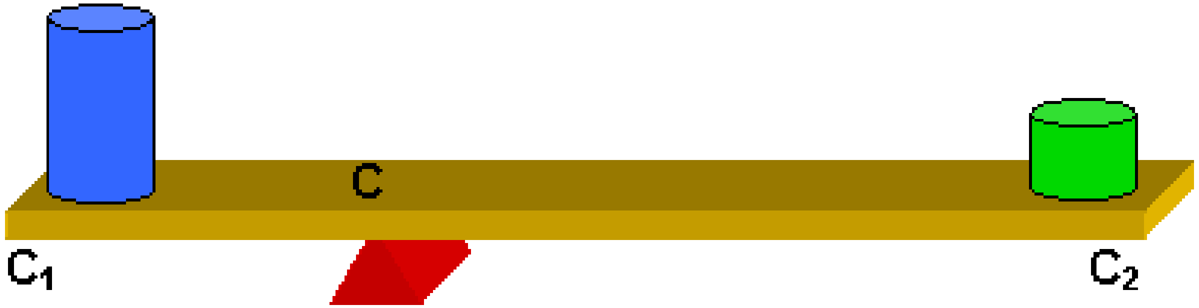
The lever rule can be explained by considering a simple balance.

The composition of the alloy is represented by the fulcrum C, and the compositions of the two phases by the ends of a bar (C_1 and C_2).

Each phase are determined by the weights needed to balance the system.

Fraction of phase 1 = $(C_2 - C) / (C_2 - C_1)$

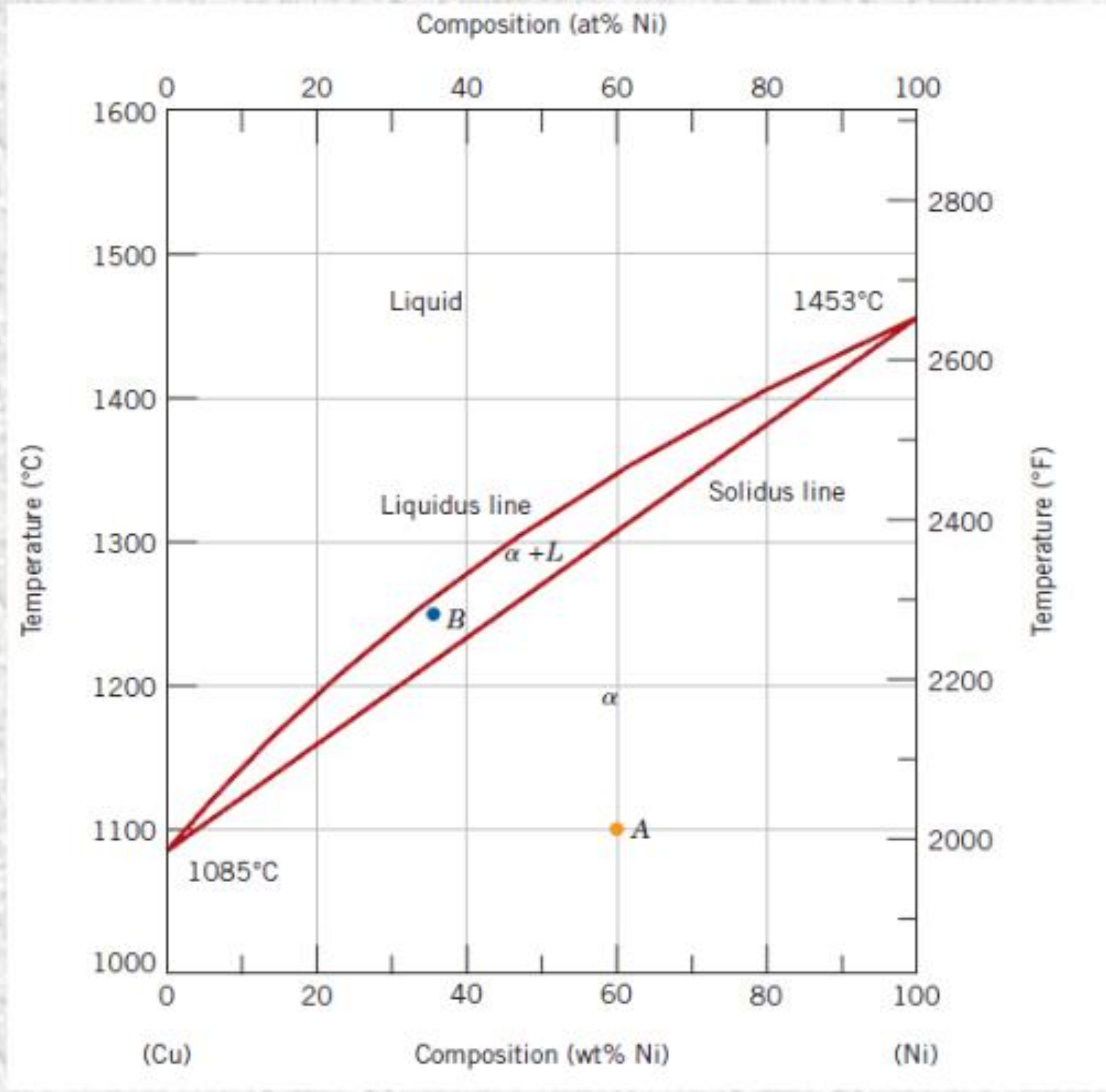
Fraction of phase 2 = $(C - C_1) / (C_2 - C_1)$.



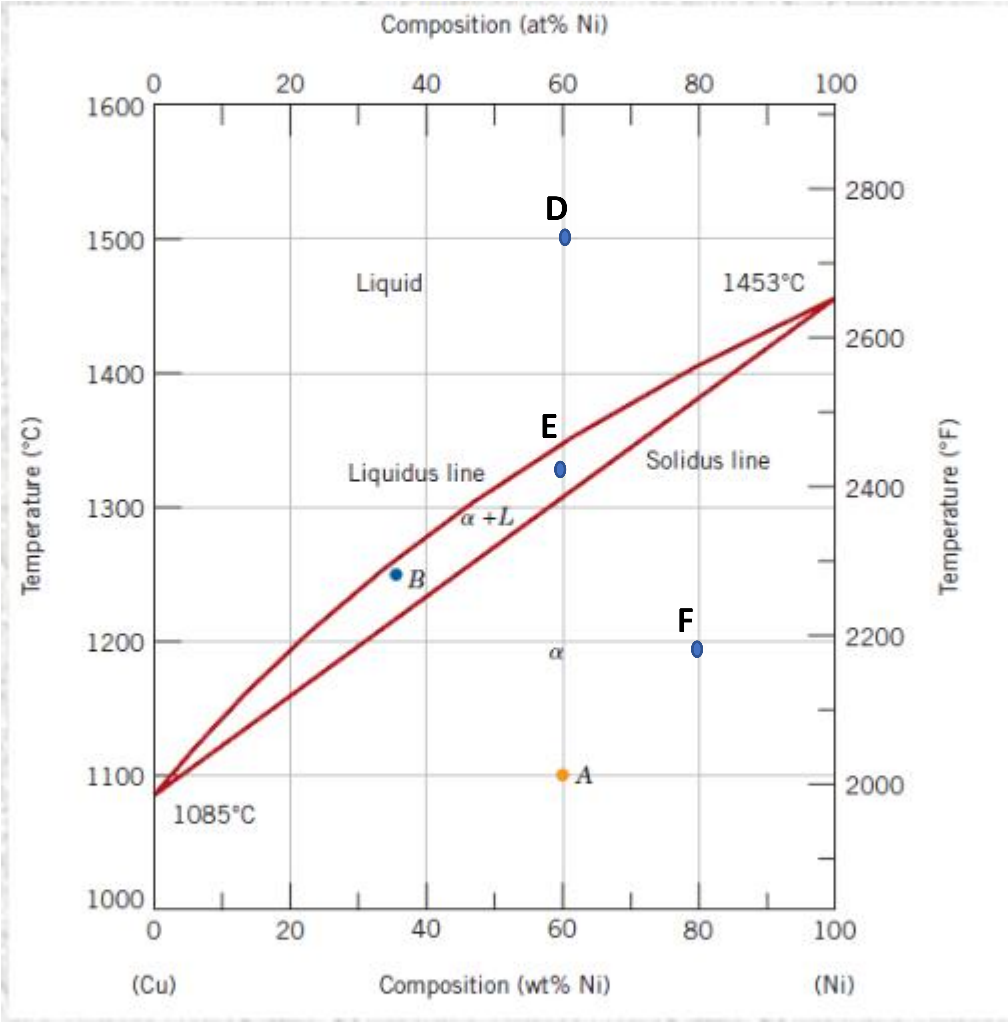
Exercise 1: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **complete solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

Exercise 2: determine the phases present, their composition and relative amounts in the point A, B and C indicated in the phase diagram. Schematize the microstructure in each point.

C

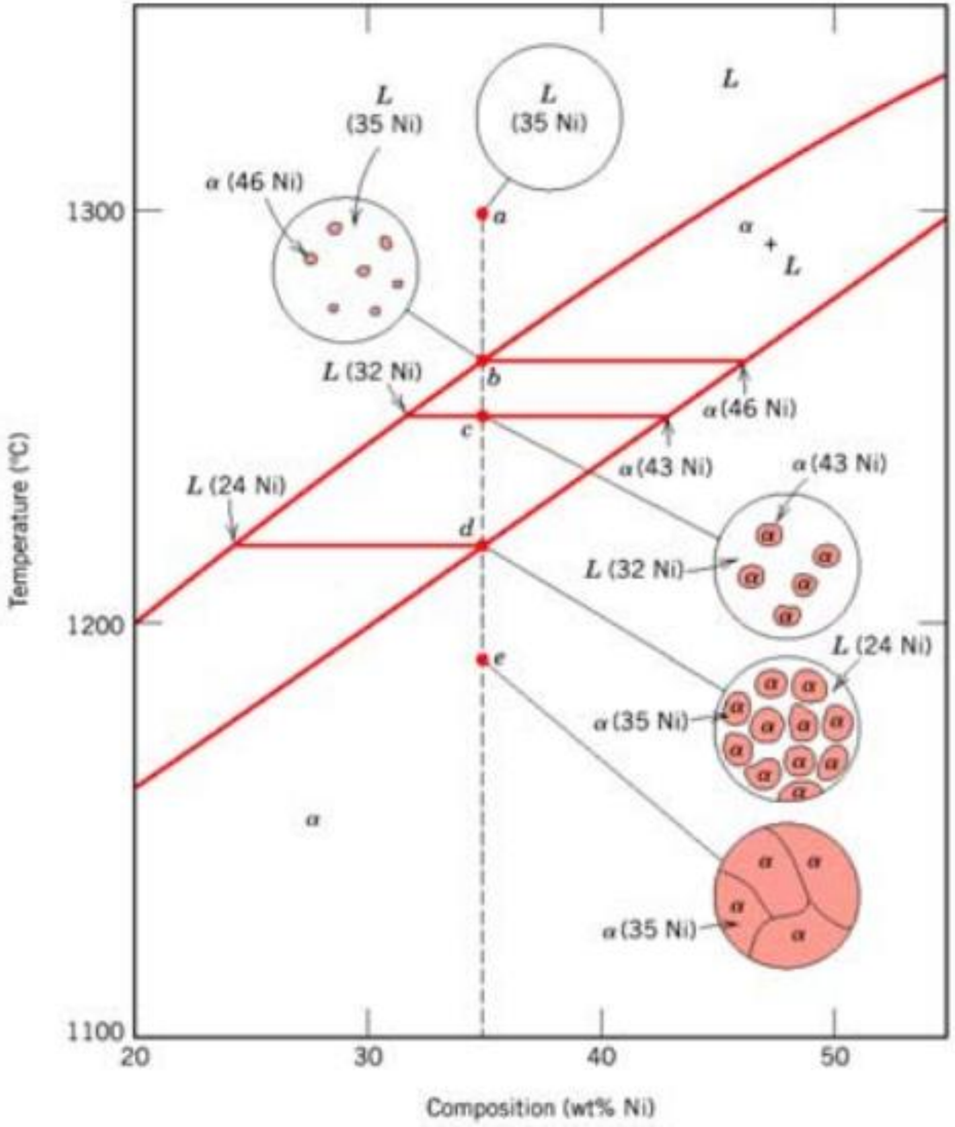


Exercise 3: determine the phase present, their composition and relative amounts in the point D, E and F indicated in the phase diagram. Schematize the microstructure in each point.



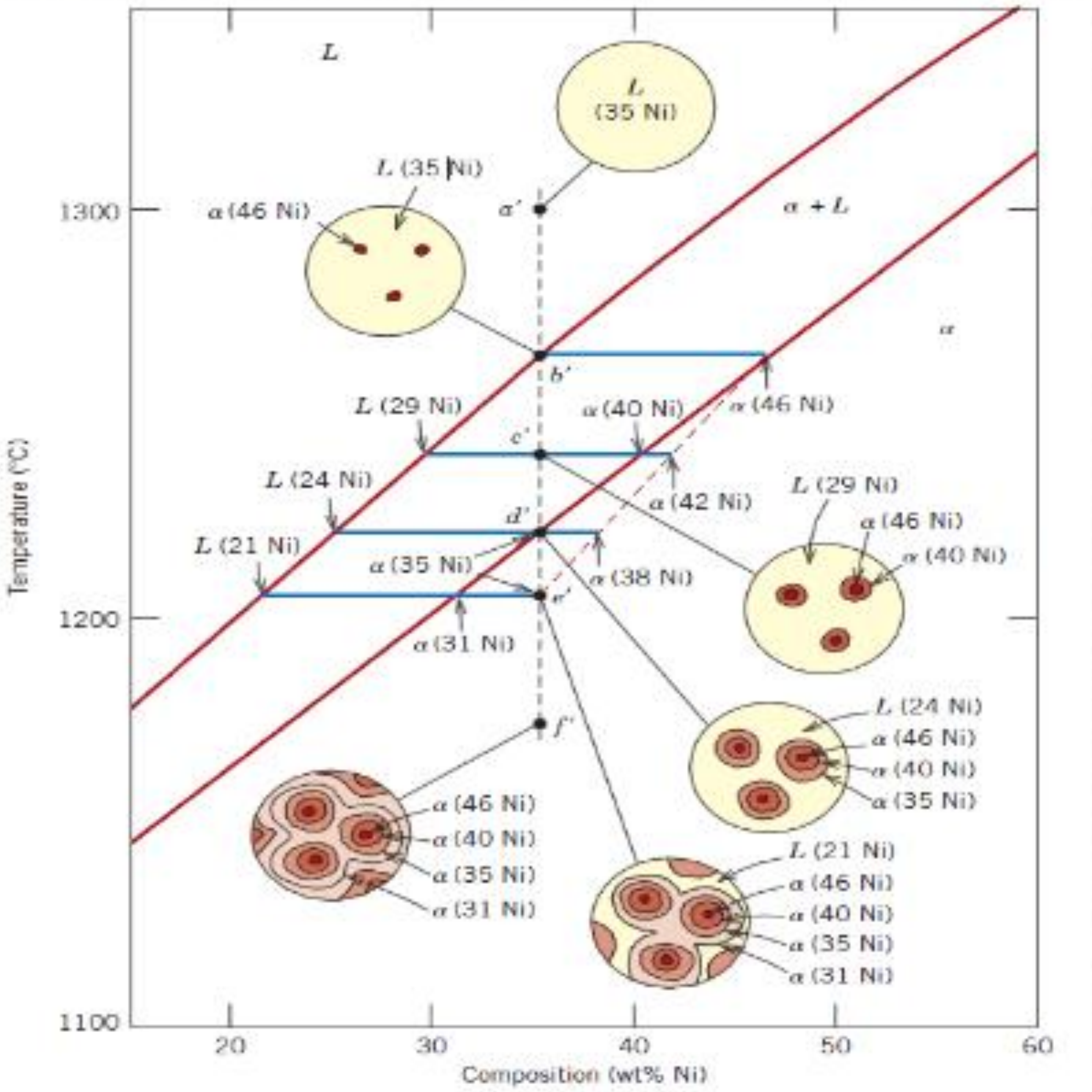
Development of microstructure in isomorphous alloys
Equilibrium (very slow) cooling

[redacted]

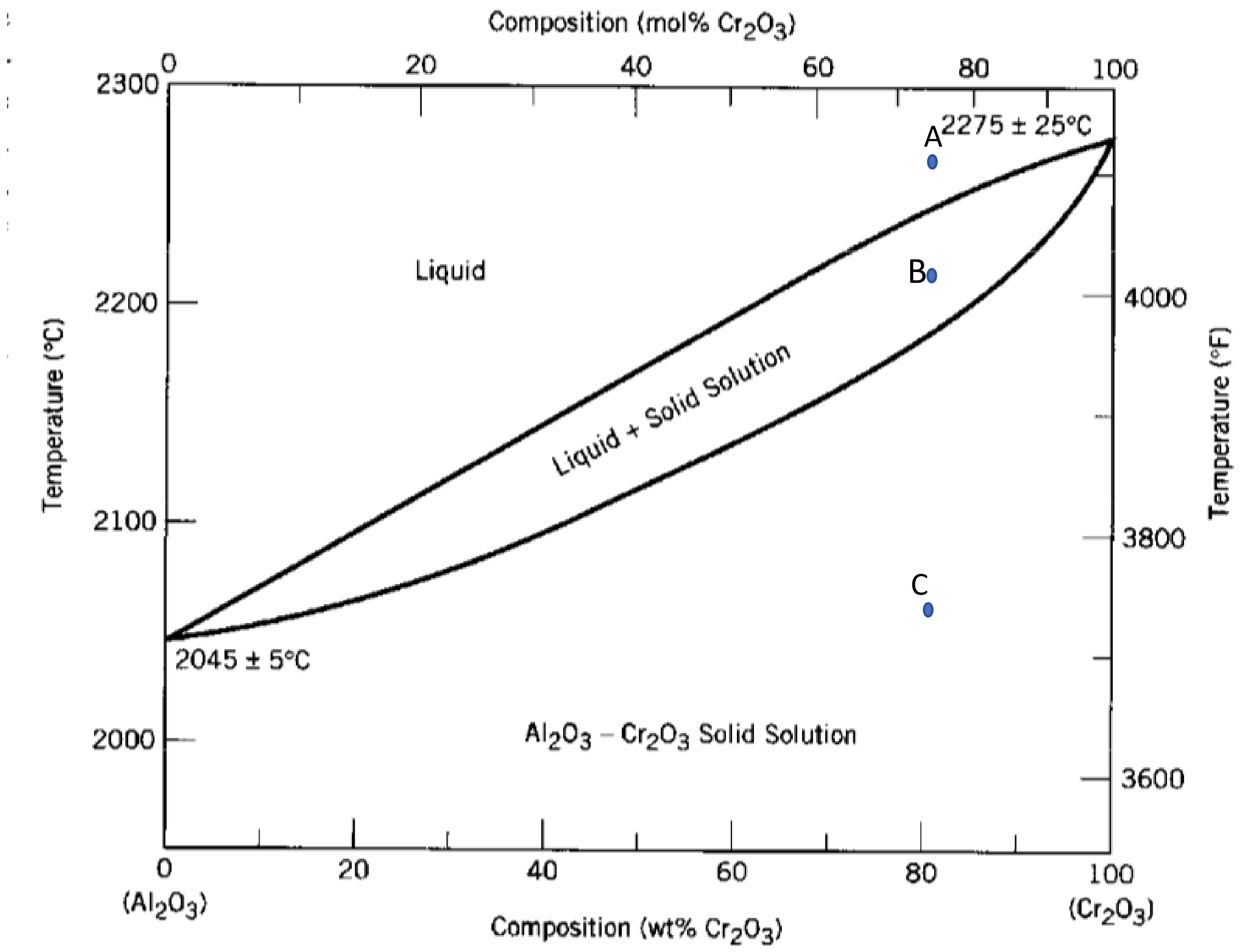


Non-Equilibrium Cooling

For Large cooling rate there is not enough time for diffusion!

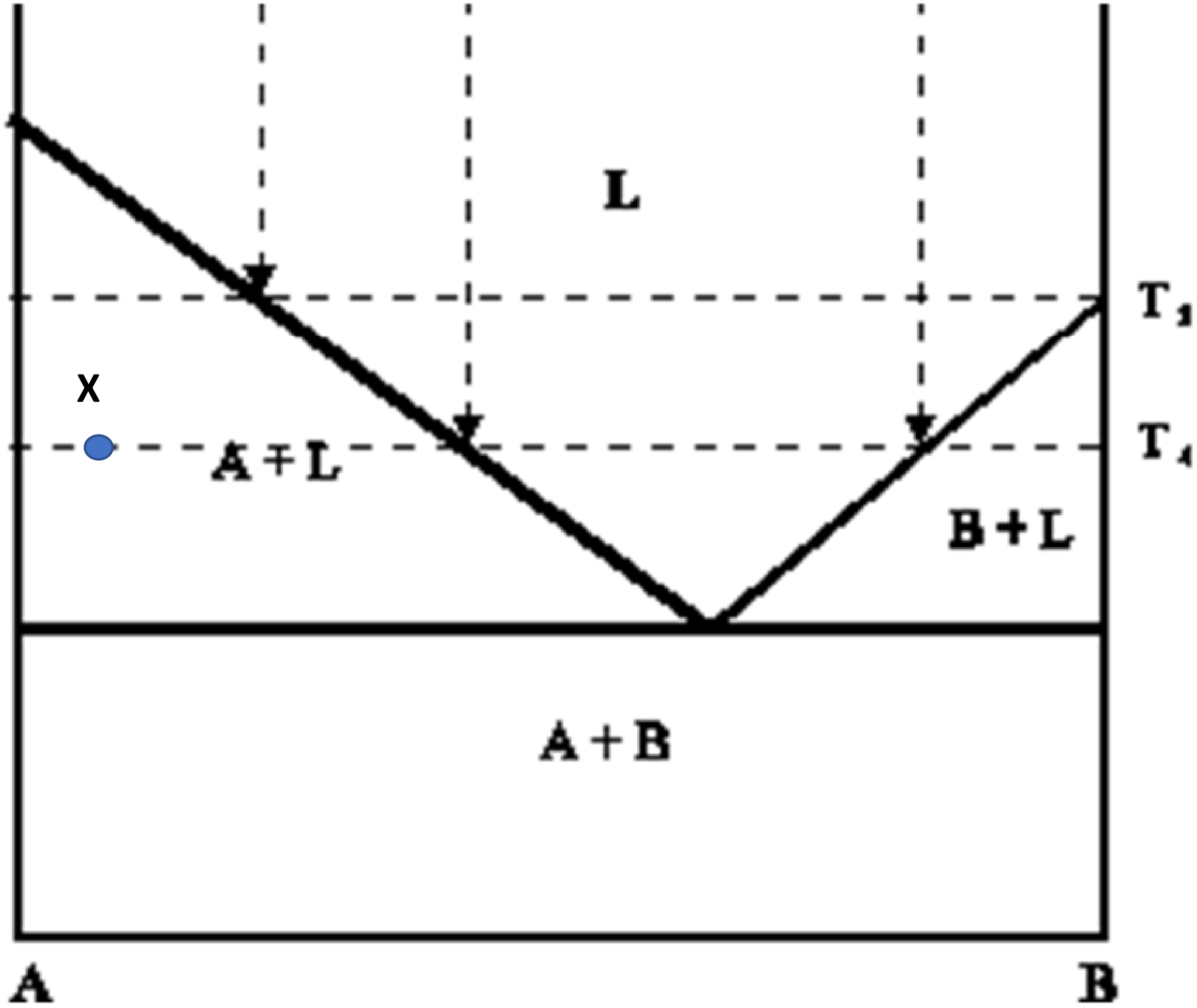


Exercise 4: determine the phase present, their composition and relative amounts in the point A, B and C indicated in the phase diagram.



Exercise 5: Draw a phase diagram (for generic components A and B), with **complete solubility in the liquid state** and **no solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

Exercise 6: determine the phases present, their compositions and relative amounts in the point X.



Exercise 7: determine the phases present, their compositions and relative amounts in the point X.